SQL JOIN

As the name shows, JOIN means *to combine something*. In case of SQL, JOIN means **"to combine two or more tables"**.

The SQL JOIN clause takes records from two or more tables in a database and combines it together.

ANSI standard SQL defines five types of JOIN :

1. inner join,
2. left outer join,
3. right outer join,
4. full outer join, and
5. cross join.

Why SQL JOIN is used?

If you want to access more than one table through a select statement.

If you want to combine two or more table then SQL JOIN statement is used .it combines rows of that tables in one table and one can retrieve the information by a SELECT statement.

The joining of two or more tables is based on common field between them.

SQL INNER JOIN also known as simple join is the most common type of join

How to use SQL join or SQL Inner Join?

Let an example to deploy SQL JOIN process:

1.Staff table

ID	Staff_NAME	Staff_AGE	STAFF_ADDRESS	Monthley_Package
1	ARYAN	22	MUMBAI	18000
2	SUSHIL	32	DELHI	20000
3	MONTY	25	MOHALI	22000
4	AMIT	20	ALLAHABAD	12000

Payment Table:

Payment_ID	DATE	Staff_ID	AMOUNT
101	30/12/2009	1	3000.00
102	22/02/2010	3	2500.00
103	23/02/2010	4	3500.00

So if you follow this JOIN statement to join these two tables ?

```
SELECT Staff_ID, Staff_NAME, Staff_AGE, AMOUNT
FROM STAFF s, PAYMENT p
WHERE s.ID =p.STAFF_ID;
```

This will produce the result like this:

STAFF_ID	NAME	Staff_AGE	AMOUNT
3	MONTY	25	2500
1	ARYAN	22	3000
4	AMIT	25	3500
1	ARYAN	22	3000

SQL OUTER JOIN

In the SQL outer JOIN all the content of the both tables are integrated together either they are matched or not.

Outer join of two types:

1.Left outer join (also known as left join): this join returns all the rows from left table combine with the matching rows of the right table. If you get no matching in the right table it returns NULL values.

2.Right outer join (also known as right join): this join returns all the rows from right table are combined with the matching rows of left table .If you get no column matching in the left table .it returns null value.

SQL LEFT JOIN

BASIC SYNTAX FOR LEFT JOIN:

1. **SELECT** table1.column1, table2.column2....
2. **FROM** table1
3. LEFTJOIN table2
4. **ON** table1.column_field = table2.column_field;

let us take two tables in this example to elaborate all the things:

Customer Table:

ID	NAME	AGE	SALARY
1	ARYAN	51	56000
2	AROHI	21	25000
3	VINEET	24	31000
4	AJEET	23	32000
5	RAVI	23	42000

Order Table:

O_ID	DATE	CUSTOMER_ID	AMOUNT
001	20-01-2012	2	3000
002	12-02-2012	2	2000
003	22-03-2012	3	4000
004	11-04-2012	4	5000

join these two tables with LEFT JOIN:

```
1. SELECT ID, NAME, AMOUNT,DATE
2. FROM CUSTOMER
3. LEFT JOIN ORDER
4. ON CUSTOMER.ID = ORDER.CUSTOMER_ID;
```

This will produce the following result:

ID	NAME	AMOUNT	DATE
1	ARYAN	NULL	NULL
2	AROHI	3000	20-01-2012
2	AROHI	2000	12-02-2012
3	VINEET	4000	22-03-2012
4	AJEET	5000	11-04-2012
5	RAVI	NULL	NULL

SQL RIGHT JOIN

The SQL right join returns all the values from the rows of right table. It also includes the matched values from left table but if there is no matching in both tables, it returns NULL.

Basic syntax for right join:

```
1. SELECT table1.column1, table2.column2.....
2. FROM table1
3. RIGHT JOIN table2
4. ON table1.column_field = table2.column_field;
```

let us take an example with 2 tables table1 is CUSTOMERS table and table2 is ORDERS table.

Customer Table:

ID	NAME	AGE	SALARY
1	ARYAN	51	56000
2	AROHI	21	25000
3	VINEET	24	31000
4	AJEET	23	32000
5	RAVI	23	42000

Order Table:

DATE	O_ID	CUSTOMER_ID	AMOUNT
20-01-2012	001	2	3000
12-02-2012	002	2	2000
22-03-2012	003	3	4000
11-04-2012	004	4	5000

Here we will join these two tables with SQL RIGHT JOIN:

```
1. SELECT ID,NAME,AMOUNT,DATE
2. FROM CUSTOMER
3. RIGHT JOIN ORDER
4. ON CUSTOMER.ID = ORDER.CUSTOMER_ID;
```

ID	NAME	AMOUNT	DATE
2	AROHI	3000	20-01-2012
2	AROHI	2000	12-02-2012
3	VINEET	4000	22-03-2012
4	AJEET	5000	11-04-2012

SQL FULL JOIN

The SQL full join is the result of combination of both left and right outer join and the join tables have all the records from both tables. It puts NULL on the place of matches not found.

SQL full outer join and SQL join are same. generally it is known as SQL FULL JOIN.

SQL full outer join:

What is SQL full outer join?

SQL full outer join is used to combine the result of both left and right outer join and returns all rows (don't care its matched or unmatched) from the both participating tables.

Syntax for full outer join:

```
SELECT *
FROM table1
FULL OUTER JOIN table2
ON table1.column_name = table2.column_name;
```

Note:here table1 and table2 are the name of the tables participating in joining and column_name is the column of the participating tables.

Let us take two tables to demonstrate full outer join:

table_A

A	M
1	m
2	n
4	o

table_B

A	N
2	p
3	q
5	r

Resulting table

A	M	A	N
2	n	2	p
1	m	-	-
4	o	-	-
-	-	3	q
-	-	5	r

Because this is a full outer join so all rows (both matching and non-matching) from both tables are included in the output. Here only one row of output displays values in all columns because there is only one match between table_A and table_B.

SQL Cross Join

- Join operation in SQL is used to combine multiple tables together into a single table.
- If we use the cross join to combine two different tables, then we will get the Cartesian product of the sets of rows from the joined table. When each row of the first table is combined with each row from the second table, it is known as Cartesian join or cross join.
- After performing the cross join operation, the total number of rows present in the final table will be equal to the product of the number of rows present in table 1 and the number of rows present in table 2.
- **For example:**
  If there are two records in table 1 and three records in table 2, then after performing cross join operation, we will get six records in the final table.
- Let us take a look at the syntax of writing a query to perform the cross join operation in SQL.

Syntax

```
SELECT TableName1.columnName1, TableName2.columnName2 FROM TableName1 CROSS JOIN TableName2 ON TableName1.ColumnName = TableName2.ColumnName;
```

Now let us see take a deeper dive into the cross join in SQL with the help of examples. All the queries in the examples will be written using the MySQL database.

Consider we have the following tables with the given data:

Example 1:

Write a query to perform the cross join operation considering the MatchScore table as the left table and the Departments table as the right table.

Table 1: MatchScore

Player	Department_id	Goals
Franklin	1	2
Alan	1	3
Priyanka	2	2
Rajesh	3	5

Table 2: Department

Department_id	Department_name
1	IT
2	HR
3	Marketing

```
SELECT * FROM MatchScore CROSS JOIN Departments;
```

We have used the SELECT command with the asterisk to retrieve all the columns present in the MatchScore and Departments table. Then we have used the CROSS JOIN keyword to perform the cross join operation on the MatchScore and Departments table. Since there are 4 records in the MatchScore and 3 records in the Departments table, after performing the cross join operation, we will get 12 rows.

After executing this query, you will find the following result:

Player	Department_id	Goals	Depatment_id	Department_name
Franklin	1	2	1	IT
Alan	1	3	1	IT
Priyanka	2	2	1	IT
Rajesh	3	5	1	IT
Franklin	1	2	2	HR
Alan	1	3	2	HR
Priyanka	2	2	2	HR
Rajesh	3	5	2	HR
Franklin	1	2	3	Marketing
Alan	1	3	3	Marketing
Priyanka	2	2	3	Marketing
Rajesh	3	5	3	Marketing

ach row from the MatchScore table is combined with each row of the Departments table. Since there are four records in the MatchScore and three records in the Departments table, we have got 12 rows in the final table after performing the cross join operation.

Conclusion:-

Different Types of SQL JOINs

Here are the different types of the JOINs in SQL:

- `(INNER) JOIN`: Returns records that have matching values in both tables
- `LEFT (OUTER) JOIN`: Returns all records from the left table, and the matched records from the right table
- `RIGHT (OUTER) JOIN`: Returns all records from the right table, and the matched records from the left table.
- `FULL (OUTER) JOIN`: Returns all records when there is a match in either left or right table

